Elizabeth Nemeti

Doctoral Candidate
Emory University

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Research Interests

Mobile Health: Ecological momentary assessment and speech processing for clinical applications.
Digital Biomarkers: Passive sensing and voice-based monitoring for mental health outcomes.
Fairness in AI: Responsible machine learning with attention to bias and equity in healthcare.

Education

Emory University, Atlanta, GA, USA — Aug 2022 - *Expected:* 2028
Ph.D. Candidate, Departments of Computer Science & Biomedical Informatics
M.Sc. in Computer Science (interim) — Jul 2025
Advisor: Prof. Selen Bozkurt
Emory Women in Natural Sciences Fellow

University of Bristol, Bristol, UK — Aug 2019 - Jul 2022
B.Sc. School of Biological Sciences
Magna Cum Laude, First Division Second-Class Honours (2:1)
Advisor: Prof. Tom Williams
Capstone: Identifying Protist Biodiversity in Lake Malawi Using Environmental DNA and Phylogenetic Bioinformatics

Research Experience

Graduate Research Assistant @ BOZ Lab P.I. Selen Bozkurt, Emory University Since May 2025	**Mental Health Prediction from Voice Diaries:** Built an end-to-end ML pipeline to identify which voice and language features (acoustic, linguistic, and multimodal) best predict depression from naturalistic daily voice diaries collected via Fabla.
Graduate R.A. @ CAV Lab P.I. Emily Wally, Emory University May - Aug 2025	**Neurodivergent-Inclusive Visualization:** Conducted a scoping review examining visualization practices that support neurodivergent users, covering concept origination, search strategy design, and corpus curation.
Graduate R.A. @ Rathore Lab P.I. Saima Rathore, Emory University Jan - May 2025	**Proteomic Analysis of Alzheimer's Pathology:** Ran differential expression analysis and integrated CSF proteomics with PET imaging to identify early AD biomarkers.
Graduate R.A. @ Next-Gen Maternal-Infant Informatics Lab P.I. Yun Wang, Emory University Aug - Dec 2024	**Fetal MRI Subcortical Segmentation:** Adapted the MAPSeg deep learning pipeline to extend subcortical brain segmentation from infant to fetal MRI for prenatal diagnostic applications.
Graduate R.A. @ Neuroinformatics & Intelligent Systems Lab P.I. Babak Mahmoudi, Emory University Dec 2023 - Jan 2025	**Automated GI Motility Quantification via Stomach MRI**: Developed an automated pipeline integrating UNet segmentation, synthetic data generation, and motility quantification for real-time GI function analysis.
Undergraduate R.A. @ Computational Evolutionary Biology Lab P.I. Tom Williams, University of Bristol Jul 2021 - Jan 2022	**Protist Identification from Environmental DNA:** Analyzed 151,000 eDNA samples to identify novel protists, advancing taxonomic classification and comparative genomics.

Publications & Presentations

[Google Scholar] | ORCID: 0009-0009-3767-7414

Publications - Published

[MIE 2026] **Voice Diaries as Digital Biomarkers: A Scoping Review**
Elizabeth Nemeti, Aravind Venkatachalam, Tricia Park, Deanna M. Kaplan, Selen Bozkurt
*Medical Informatics Europe **(MIE)**, 2026*

Publications - Submitted for Publication

[Under Review] **Towards Neurodiversity-Informed Visualization: Four Roles of Visualization in Supporting Neurodivergent Users**
Mengyu Chen, Yanan Da, Thao Le, **Elizabeth Nemeti**, Keke Wu, Emily Wall
*IEEE Transactions on Visualization and Computer Graphics **(TVCG)**, 2026*

[Under Review] **Daily Diaries as Measurement Instruments: Linguistic and Acoustic Design Considerations for Depressive Symptom Detection**
Elizabeth Nemeti, Lillian Ward, Santiago J Alvarez Arconada, Hyeokhyen Kwon, Deanna M. Kaplan, Selen Bozkurt
*Advances in Methods and Practices in Psychological Science (**AMPPS**), 2026*

Presentations *presenting author

[SAA 2026] **Phenomenological Ambulatory Assessment from Start to Finish: Centering Participant Subjectivity in Acoustic Analysis of Daily Speech Features [Symposium Talk]**
Elizabeth Nemeti (with Gil Grunfeld, Roman Palitsky, Deanna Kaplan, Matthias Mehl)
Society for Ambulatory Assessment (SAA) 2026 Conference. August 3-5, 2026. Vienna, Austria.

[AAIC 2025] **Predicting longitudinal accumulation of tau pathology in a pre-clinical population using machine learning: Data from the A4 and LEARN Studies [Oral Presentation]**
Melika Saadati, **Elizabeth Nemeti**, Saima Rathore*
Alzheimer's Association International Conference (AAIC) 2025. [Month, dates], 2025. [City, Country]. Abstract: Alzheimer's & Dementia, 21(S1).

[DDW 2025] **An automated segmentation pipeline for quantifying gastric motility in humans using 4D cine magnetic resonance imaging [Poster]**
Elizabeth Nemeti, Sarper Yurtseven, Braden Kuo, Vitaly Napadow, Roberta Sclocco*, Babak Mahmoudi
Digestive Disease Week (DDW) 2025. [Month, dates], 2025. [City, Country]. Abstract: Gastroenterology, 168(6S).

Honors, Awards, & Fellowships

CompBio Asia Advanced Studies Workshop **(NSF-funded)**, Bangkok, Thailand 2025
2-week intensive training in AI, genomics, proteomics, structural biology, and molecular dynamics.
Emory Kharen Fulton Diversity Graduate Award Nominee 2024
Emory BMI Department Graduate Student Award for Community Service 2024
Emory Women in Natural Sciences Fellowship 2023

Teaching

Teaching Assistant, Emory University
CS 110 Computer Science Fundamentals Fall 2024 & Spring 2025
CS 312 Computing, AI, Ethics and Social Theory Fall 2023
K-12 Language Assistant, Colegio Los Olivos, Málaga, Spain Spring 2019
Taught English immersion classes across multiple grade levels (K-12) as part of TEFL certification with Meddeas, including

live-in host family tutoring.

Services

Science Communication & Public Engagement	
Co-founder & President @ Biomedical Informatics Outreach Group Emory Since May 2023	Founded and led OpenBMI, a graduate student organization advancing BMI education for underserved STEM communities through public outreach events (Atlanta Science Festival 2023, 2025), campus programming, and organizational operations including treasury, risk management, and media presence.
Science Writer @ Bare Bones-STEM Emory Since 2025	Science article writer for a graduate-level science communication initiative.
STEM Public Engagement Volunteer @ Association for Women in STEM 2023-2025	Partnered with AWIS to run educational outreach booths at science festivals and museums, showcasing international women's research achievements in BMI.
Interdisciplinary Coordinator @ Women in STEM Bristol 2020-2021	Co-led efforts to foster an inclusive STEM community by developing and coordinating educational and social events that increased student engagement and amplified the visibility of women in science.

Program Committee Member, Reviewer	
Medical Informatics Europe **(MIE)** Reviewer	2026
PNAS Nexus (co-reviewer)	2026
Computing in Cardiology **(CinC)**	2023

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